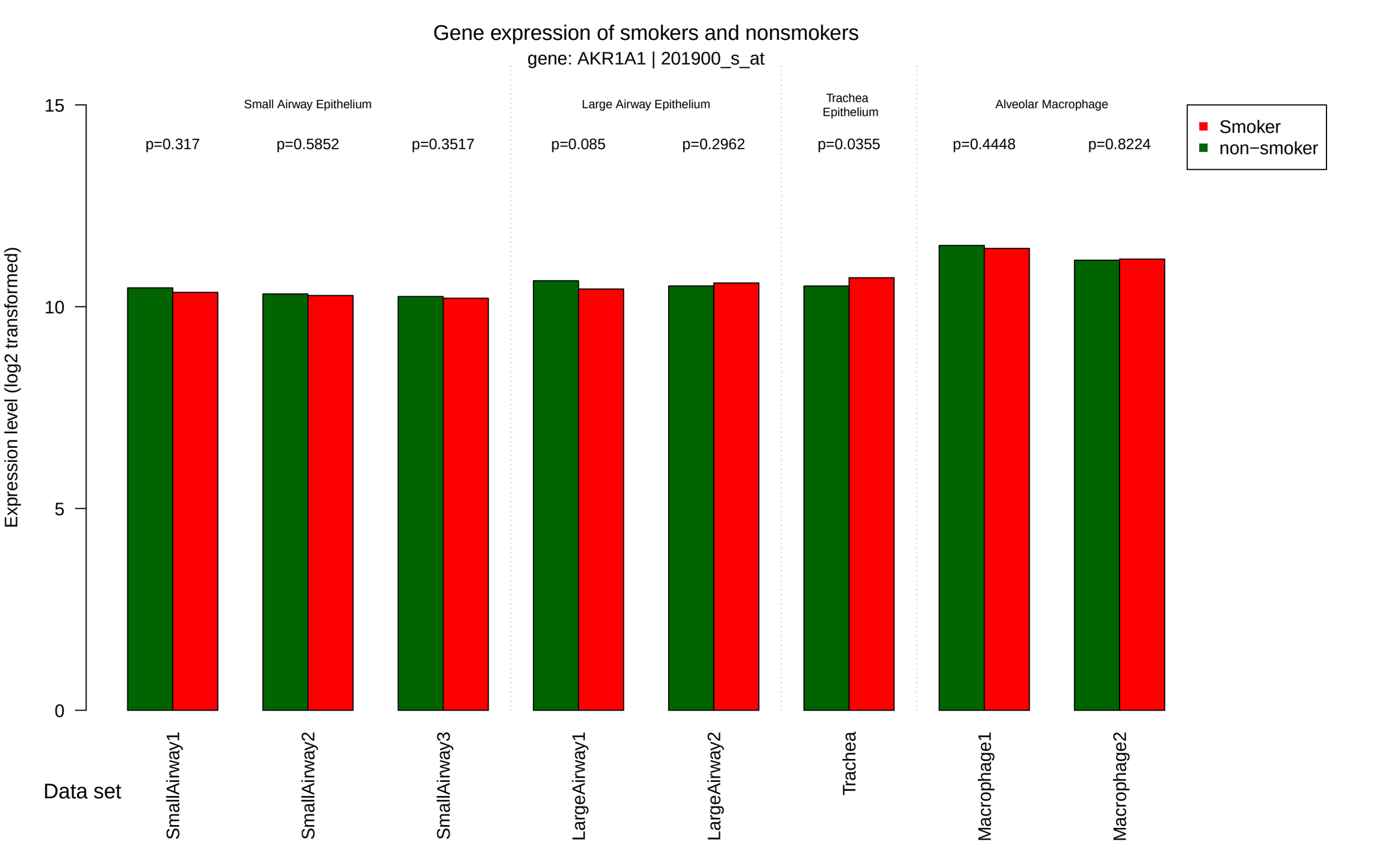
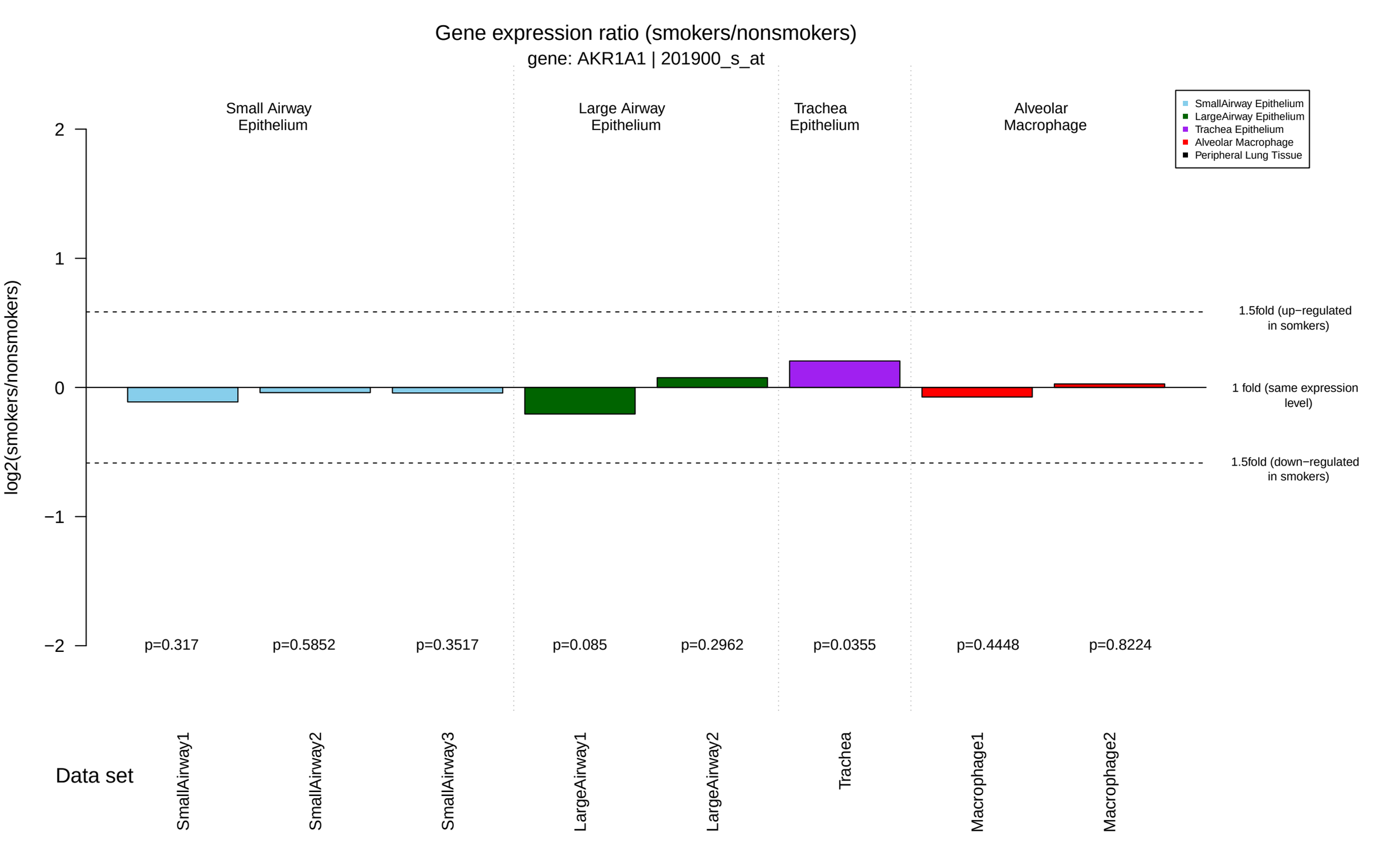
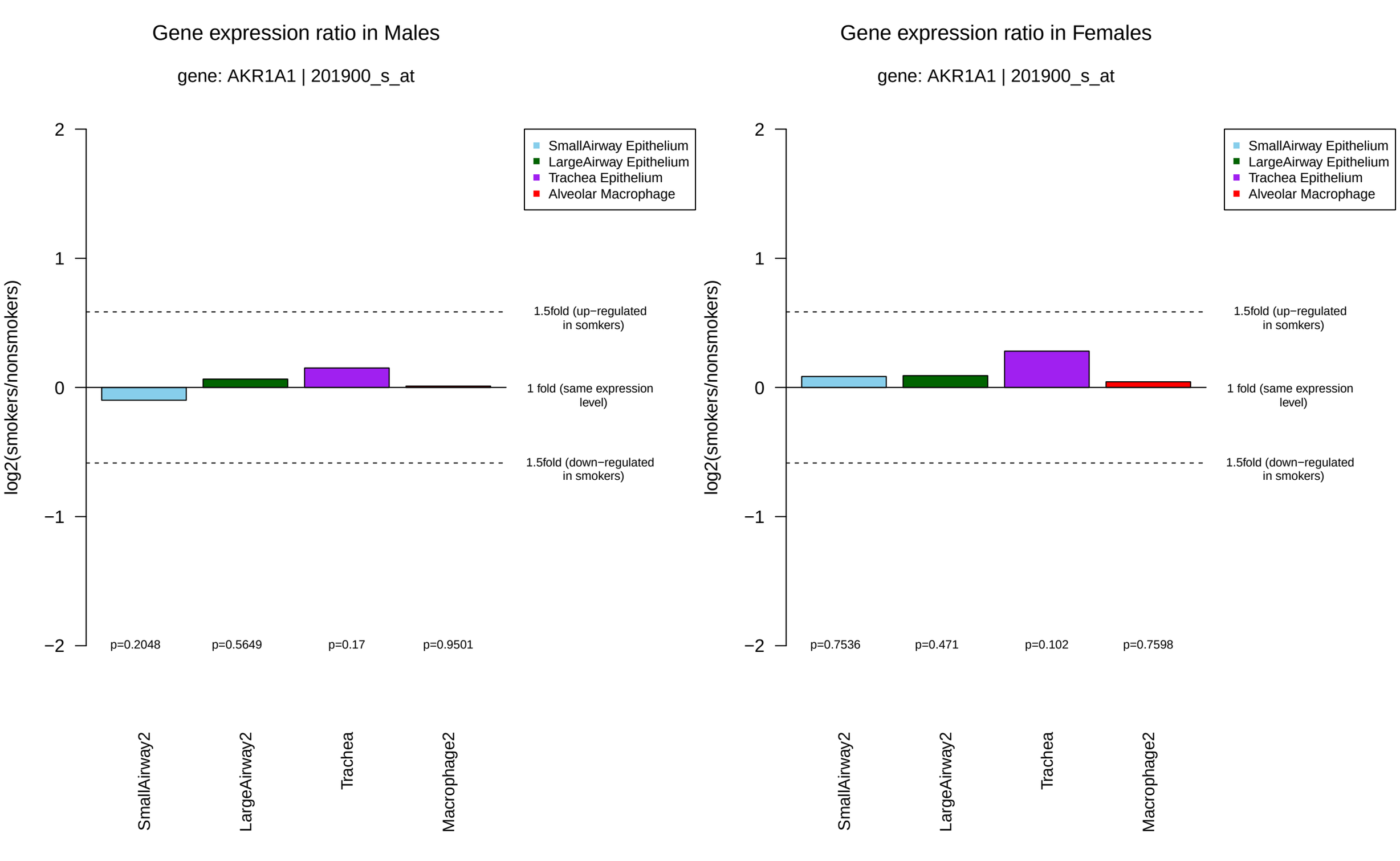
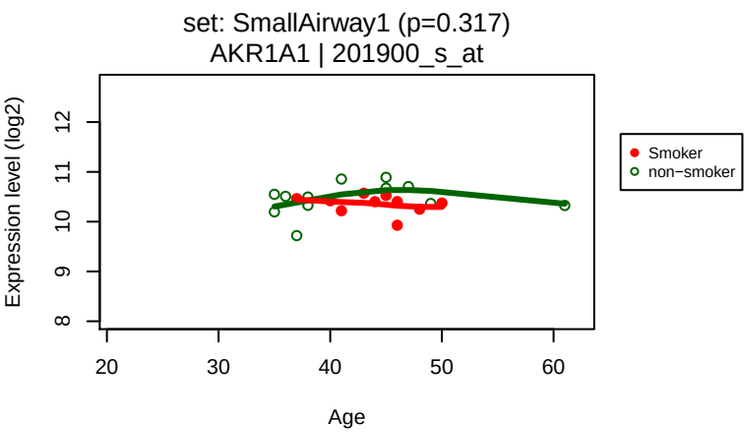
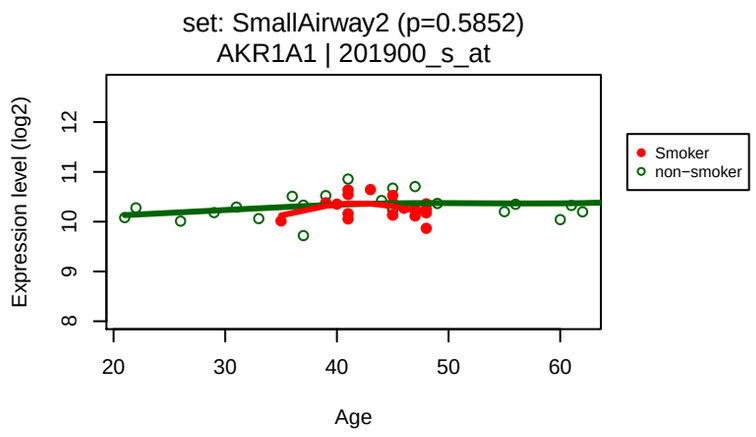
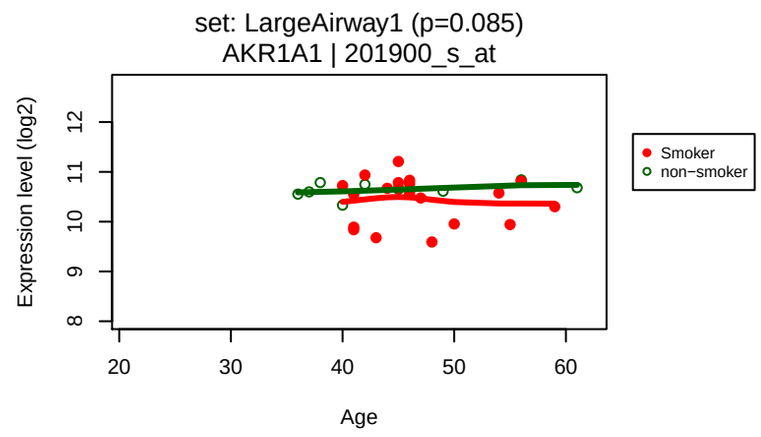

Gene expression of smokers and nonsmokers
gene: AKR1A1 | 201900_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.317
p=0.5852
p=0.3517
p=0.085
p=0.2962
p=0.0355
p=0.4448
p=0.8224
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

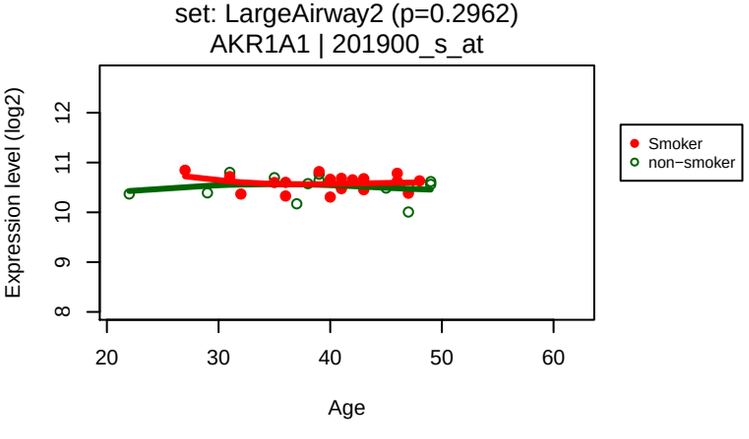

Gene expression ratio (smokers/nonsmokers)
gene: AKR1A1 | 201900_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.317
p=0.5852
p=0.3517
p=0.085
p=0.2962
p=0.0355
p=0.4448
p=0.8224
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

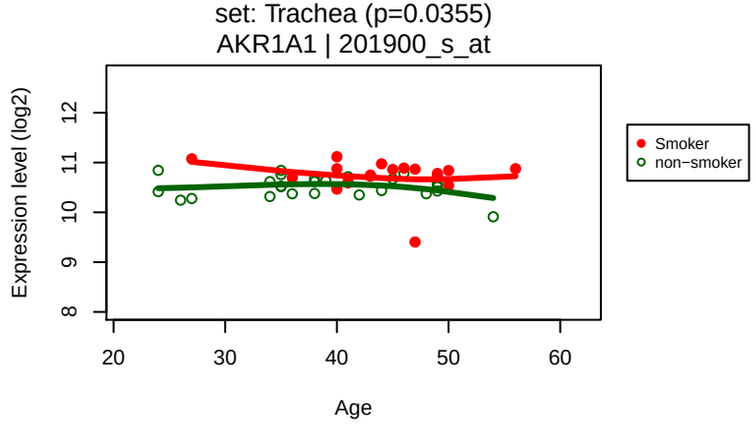

Gene expression ratio in Males
gene: AKR1A1 | 201900_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2048
p=0.5649
p=0.17
p=0.9501
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKR1A1 | 201900_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7536
p=0.471
p=0.102
p=0.7598
SmallAirway2
LargeAirway2
Trachea
Macrophage2

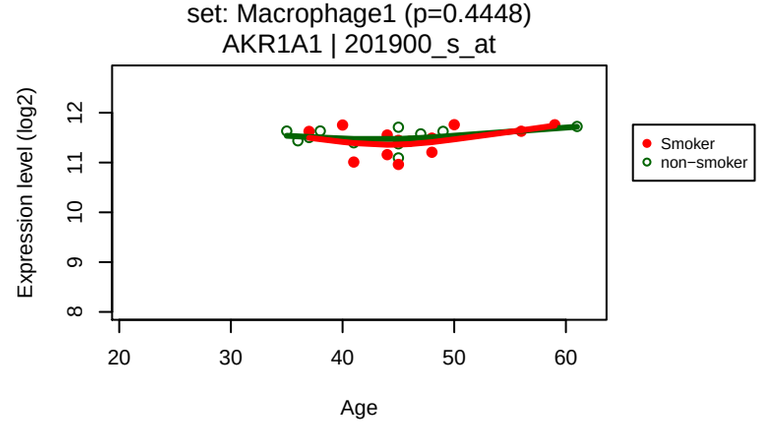

set: SmallAirway1 (p=0.317)
AKR1A1 | 201900_s_at
Expression level (log2)
Age
Smoker
non−smoker

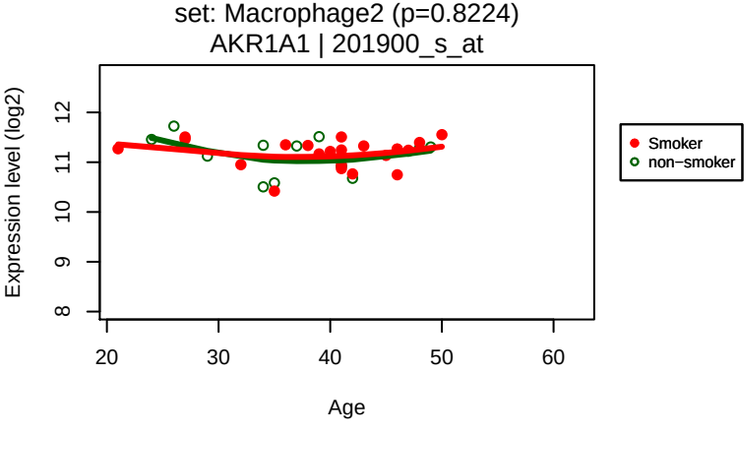

set: SmallAirway2 (p=0.5852)
AKR1A1 | 201900_s_at
Expression level (log2)
Age
Smoker
non−smoker

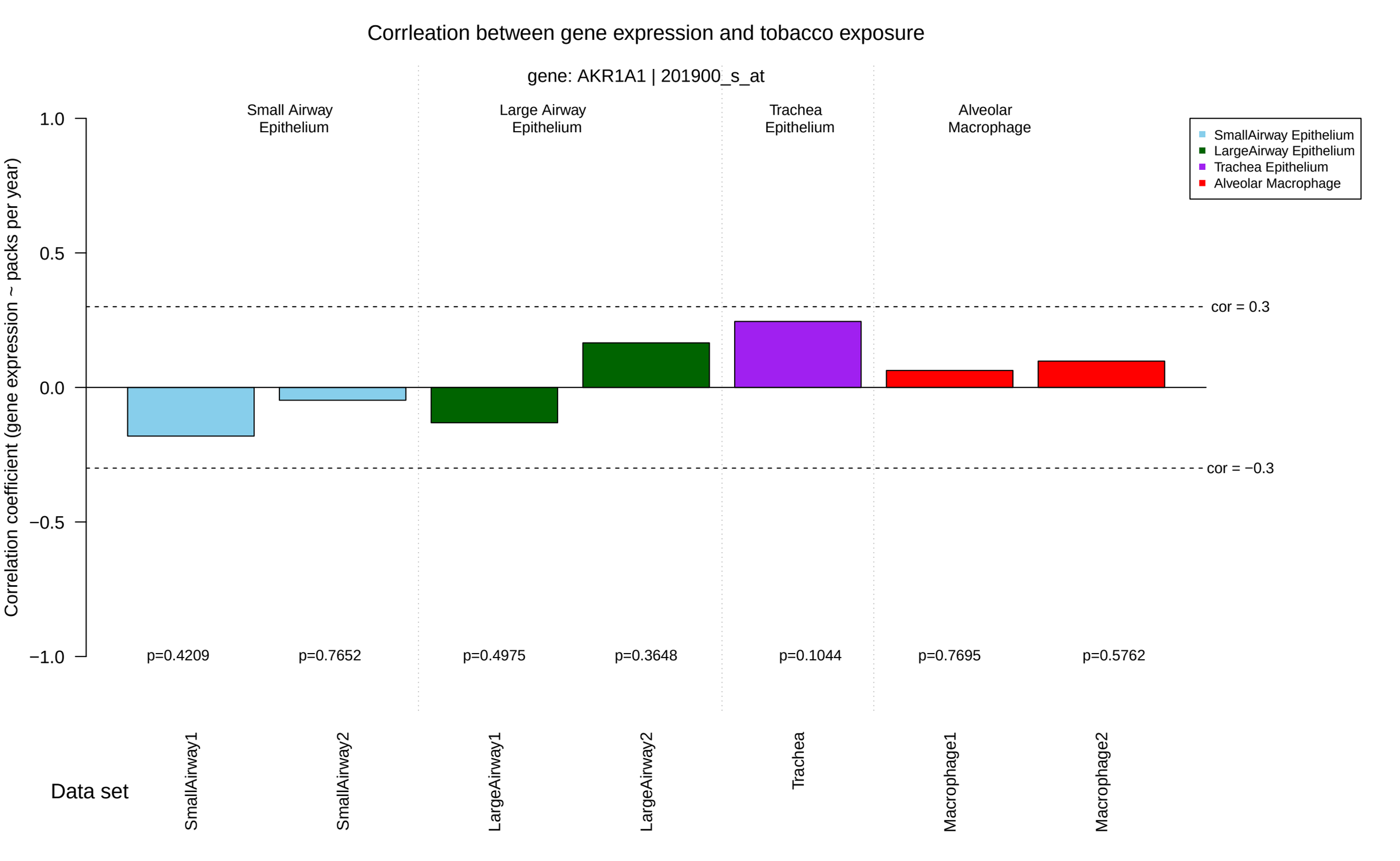

set: LargeAirway1 (p=0.085)
AKR1A1 | 201900_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2962)
AKR1A1 | 201900_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0355)
AKR1A1 | 201900_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4448)
AKR1A1 | 201900_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8224)
AKR1A1 | 201900_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKR1A1 | 201900_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4209
p=0.7652
p=0.4975
p=0.3648
p=0.1044
p=0.7695
p=0.5762
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2